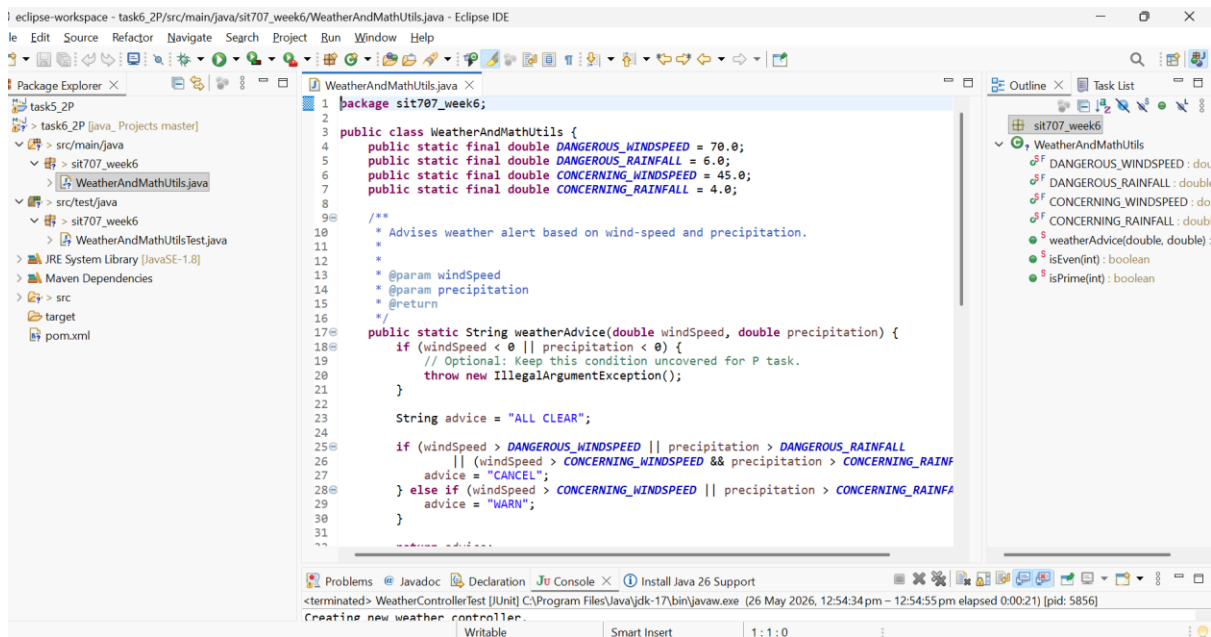


Pass Task 6: Unit test and code coverage



Successfully Downloaded the zip file extracted and then imported inside maven.

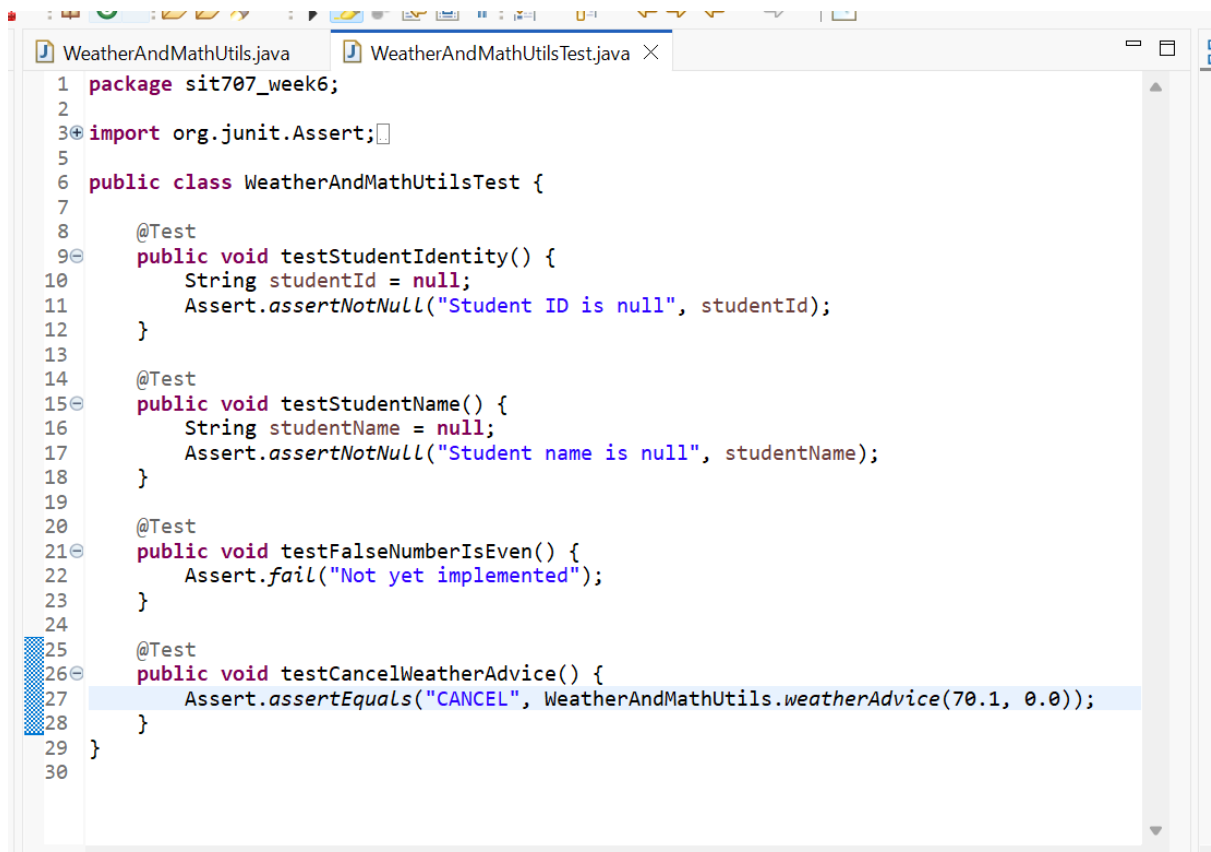
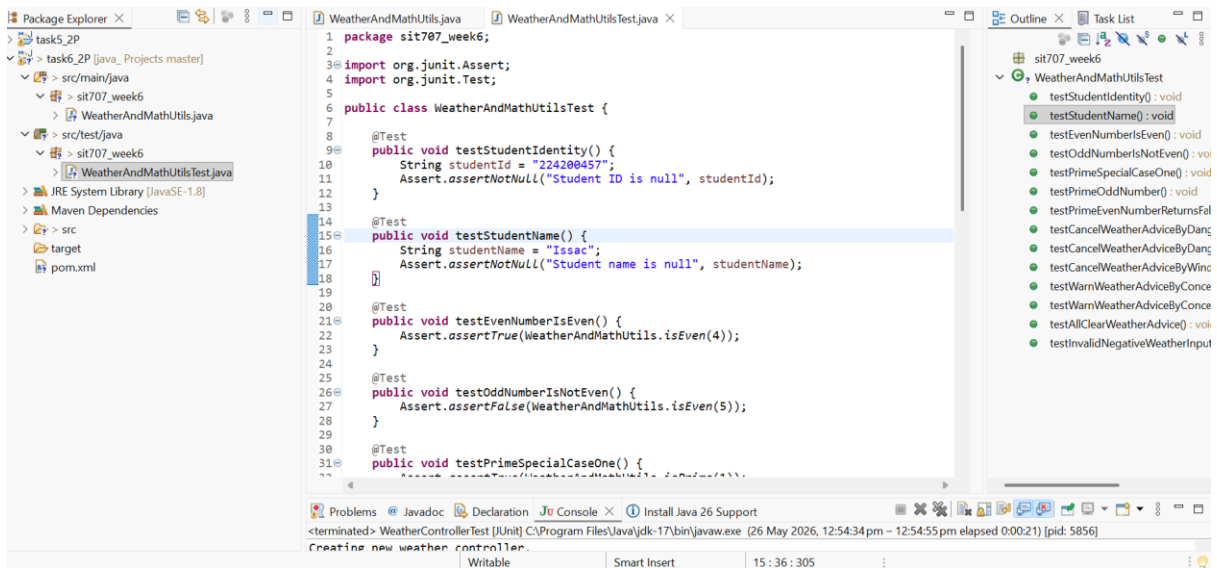
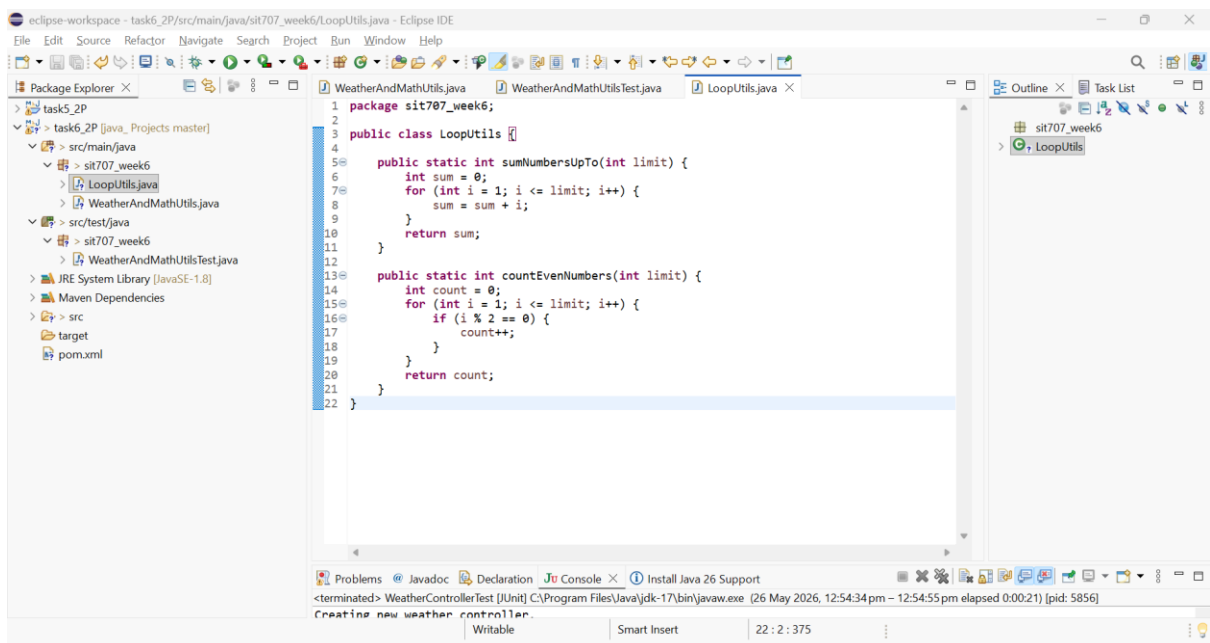


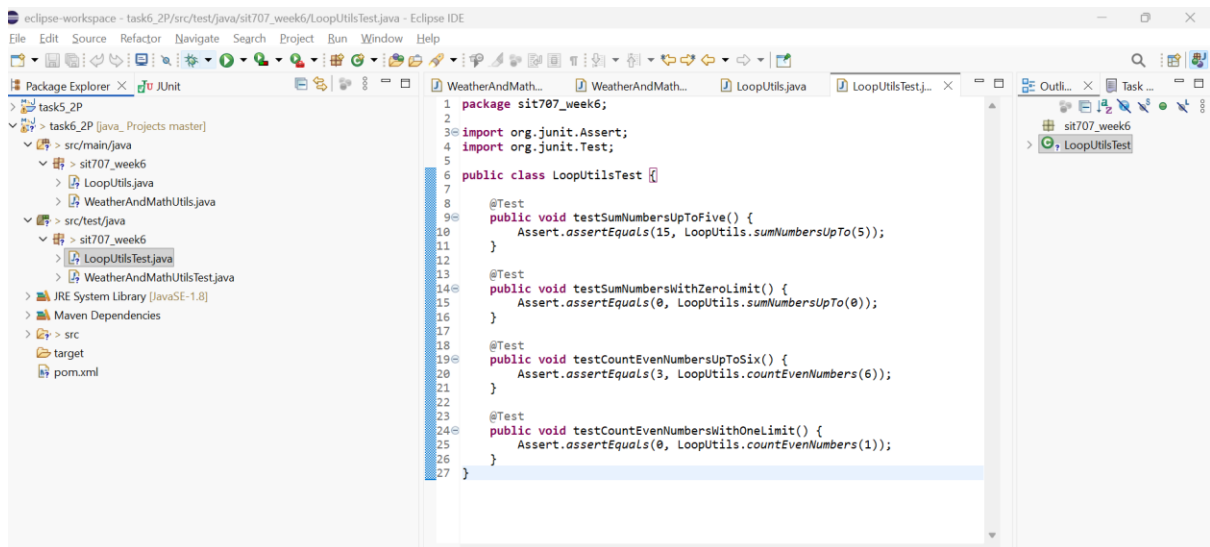
Figure 1 Old file



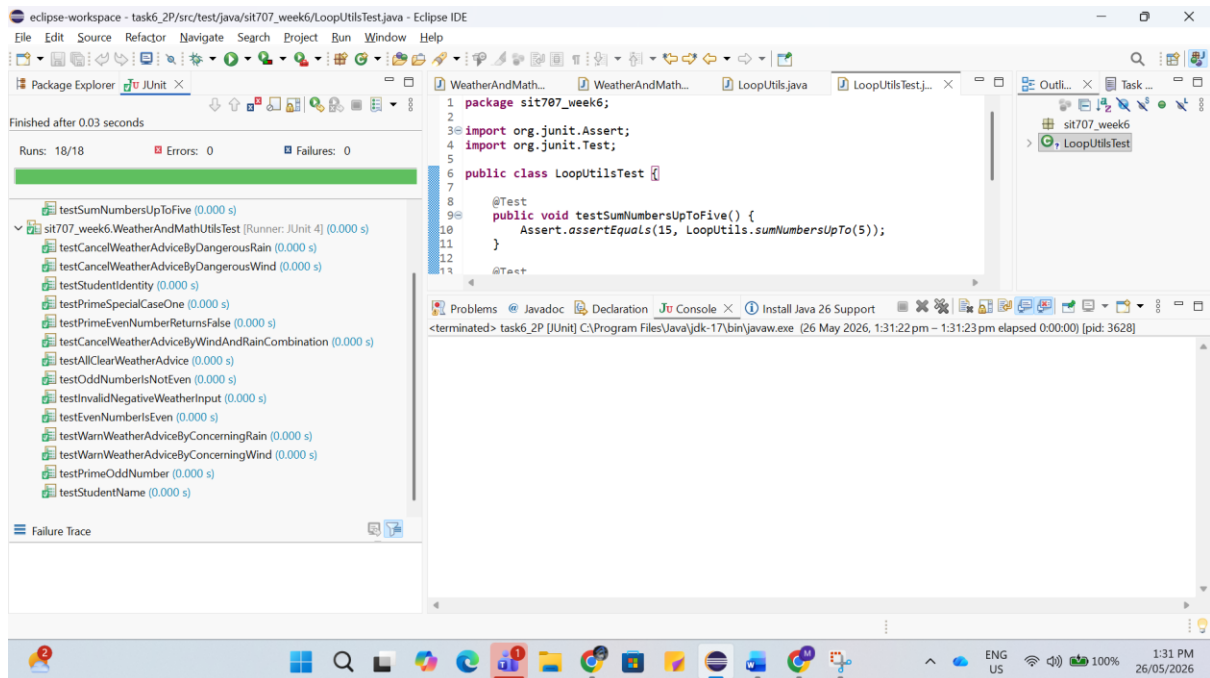
Test file updated

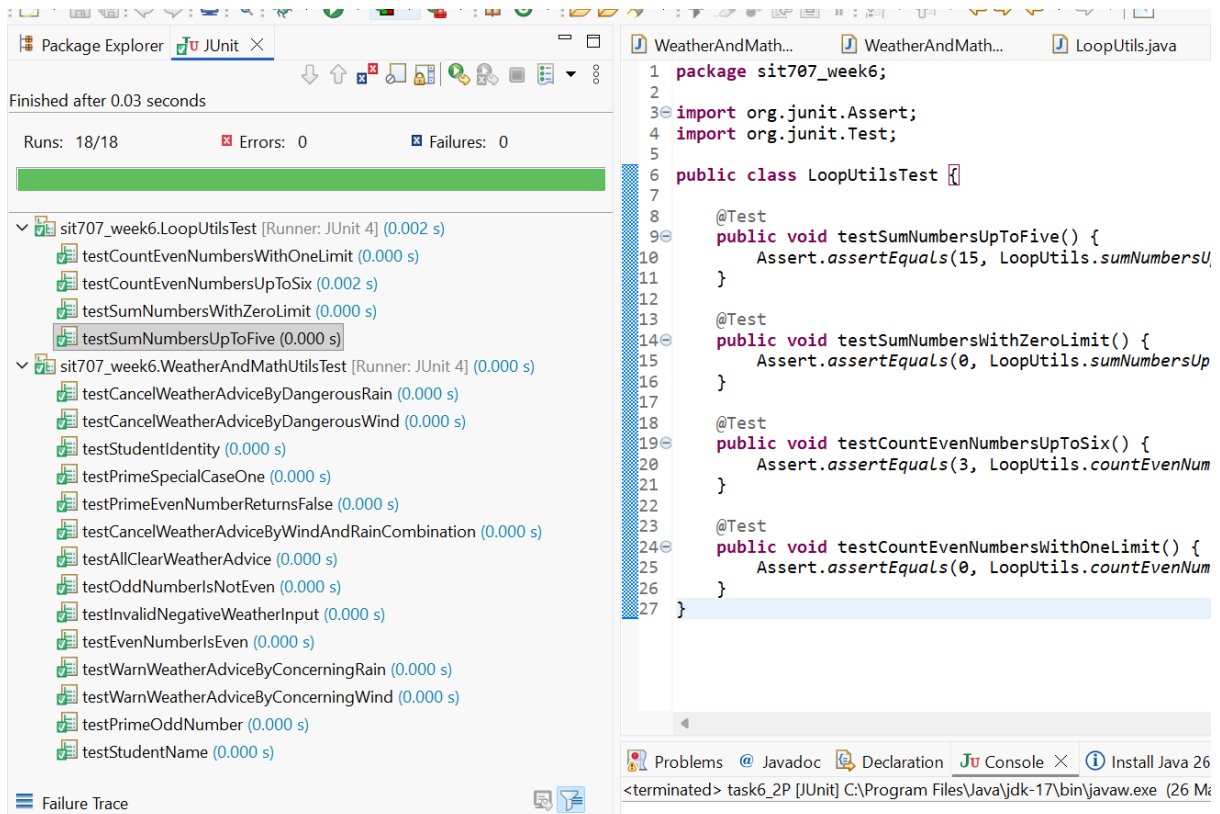


Created part b source file and updated it



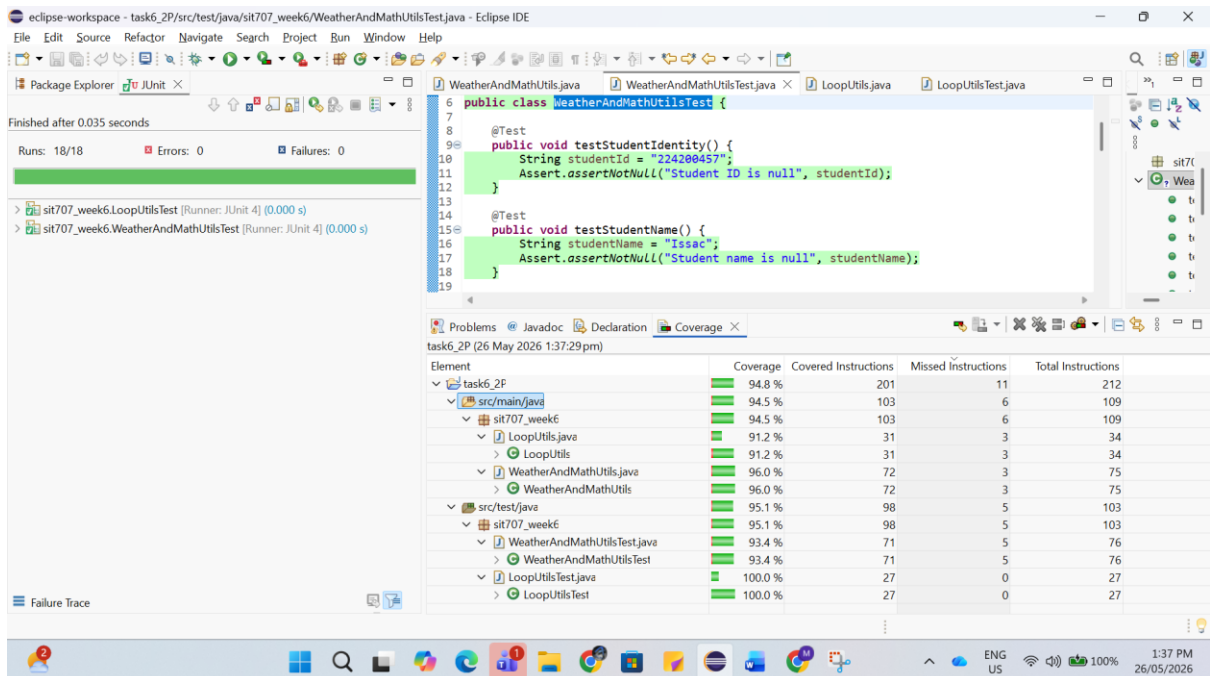
Created test for part b





Successfully ran it

And then ran coverage as Junit





LoopUtilsTest.java



WeatherAndMathUtilsTest.java



LoopUtils.java

Source code for:

Code Coverage Discussion

Code coverage measures how much of the source code is executed by unit tests. Higher code coverage helps identify untested sections of the program and improves software reliability. In this task, additional JUnit tests were created to test different weather alert conditions, even/odd number validation, prime number logic, and loop-based functions. This improved the coverage of WeatherAndMathUtils.java and LoopUtils.java to above 90%.

However, code coverage also has limitations. High coverage does not always guarantee bug-free software because some logical errors may still exist even when statements are executed. For example, branch conditions, invalid inputs, and edge cases may still require additional testing and manual analysis.

Path analysis is important in unit testing because different paths in the source code may produce different outputs. Statement coverage checks whether lines of code are executed, condition coverage checks decision outcomes such as true/false conditions, and loop coverage checks whether loops execute correctly for different iteration counts.

Github Push

The screenshot shows a GitHub repository named 'IssacSIT707'. The repository has 1 branch and 0 tags. The file list includes:

- Task_4.1_p: first commit of task4.1 (last month)
- task2_1P: task_2.1 (2 months ago)
- task3_1P: first commit (last month)
- task5_1P: Added task5_1P Maven project (1 hour ago)
- task6_1P: Added task6_1P code coverage task (now)
- week01: week01 task_1.2 (2 months ago)
- .gitignore: task_2.1 (2 months ago)
- New Text Document.txt: Added task5_1P Maven project (1 hour ago)
- welcome.md: Add welcome message to welcome.md (2 months ago)

The sidebar on the right shows repository statistics:

- About: No description, website, or topics provided.
- Activity: 0 stars, 0 watching, 0 forks.
- Releases: No releases published. [Create a new release](#)
- Packages: No packages published. [Publish your first package](#)
- Contributors: 1 contributor (Issac).